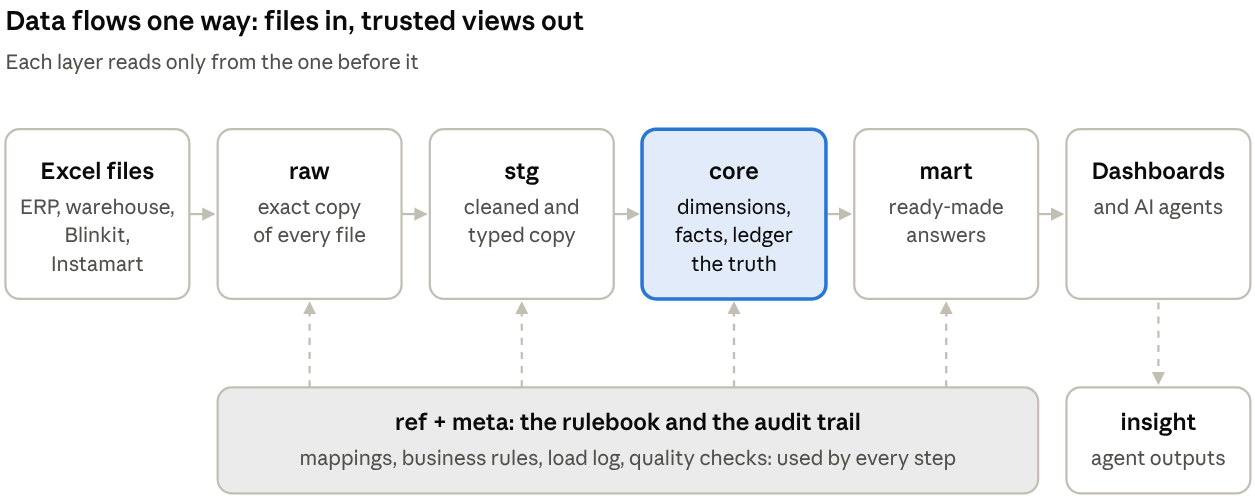
Ellementry Data Warehouse — How the Schema Works

Oct 3, 2026 · @Rajat

A plain-language guide to every table in the warehouse, what one row means, and how the tables connect. Numbers quoted come from the sample build (Apr 2025 – Sep 2026 sample files).

1. The layers



layers · files to raw, stg, core, mart, then dashboards and agents

The warehouse is organised in layers, like stations on an assembly line. Each one does one job and reads only from the one before it, so a problem can always be traced back to where it started.

Layer	What it holds	In plain words
raw	14 tables, one per kind of export, every column as text	A photocopy of each file exactly as received. Never changed, so we can always go back to the original.
stg	Cleaned copies of raw: dates are dates, numbers are numbers	The same data, tidied: spaces trimmed, text turned into proper dates and amounts. Rebuilt every run.

Layer	What it holds	In plain words
core	Dimensions, facts and the stock ledger	The single version of the truth. Business rules are applied here.
mart	10 ready-made views	Pre-built answers for dashboards and agents, such as net sales per day or stock at every location.
insight	5 tables	What the AI agents produce: forecasts, recommendations and alerts, plus the human decision on each.
ref and meta	Mappings, rules, logs	The rulebook (how each name maps, what each rule says) and the audit trail (what was loaded, when, and whether it passed its checks).

Most people only ever need **mart**. Analysts and agents that need more detail use **core**. Nobody outside the data team should read **raw**, because it contains customer personal data exactly as exported.

2. Two kinds of tables, and how they connect

Almost every table in **core** is one of two kinds.

- **Facts record events.** Something happened: an invoice line was billed, a return came back, stock moved, a PO was raised. Fact tables grow every day and hold the numbers: quantities and amounts. Their names start with `fact_`.
- **Dimensions describe things.** Who, what, where, when: the product, the location, the channel, the partner, the date. They change slowly and hold names and attributes, not running totals. Their names start with `dim_`.

Think of a fact table as a receipt book, and the dimensions as the product catalogue, the address book and the calendar. A receipt doesn't repeat the full product description; it writes a code you can look up in the catalogue.

Keys: the codes that connect them

Every dimension row has a **key**: a number such as `product_key = 1842`. A fact row stores that number instead of the product's name and details. To find the product's name, you look up the key in `dim_product`. That lookup is called a **join**.

Three conventions hold everywhere:

- **Key -1 means "Unknown".** If a file mentions a SKU or location that doesn't match anything, the row is kept and given key -1, never dropped. A data-quality check counts every -1, so nothing disappears silently.
- **Keys never change.** Once a product, location or partner has a key, it keeps it on every rebuild, so forecasts and recommendations saved by the agents always point at the right thing.
- **The original value is kept too.** Facts also store the SKU exactly as it was written in the file (`source_sku_code`), so an Unknown row can always be traced back to what the file said.

Worked example: reading one invoice line

One row in `fact_sales_line` might read:

Column	Value	What it means
bill_number	JPR26SP-S1	The ERP invoice number
bill_date_key	20260401	Look up in `dim_date`: Wednesday 1 April 2026, fiscal year 2026-27, Q1
product_key	1842	Look up in `dim_product`: SKU GSTEA2724, Glass Bottle, Continued
location_key	1	Look up in `dim_location`: Bagru warehouse
channel_key	1	Look up in `dim_channel`: Own Website (Shopify)
partner_key	4	Look up in `dim_partner`: Shopify_Customer, a D2C channel account
quantity	1	One unit sold
net_sales_amount	705.93	Rs 705.93 excluding GST (rule BR-001)

Key values here are illustrative; the meanings are exactly how the tables connect.

3. The dimensions: who, what, where, when

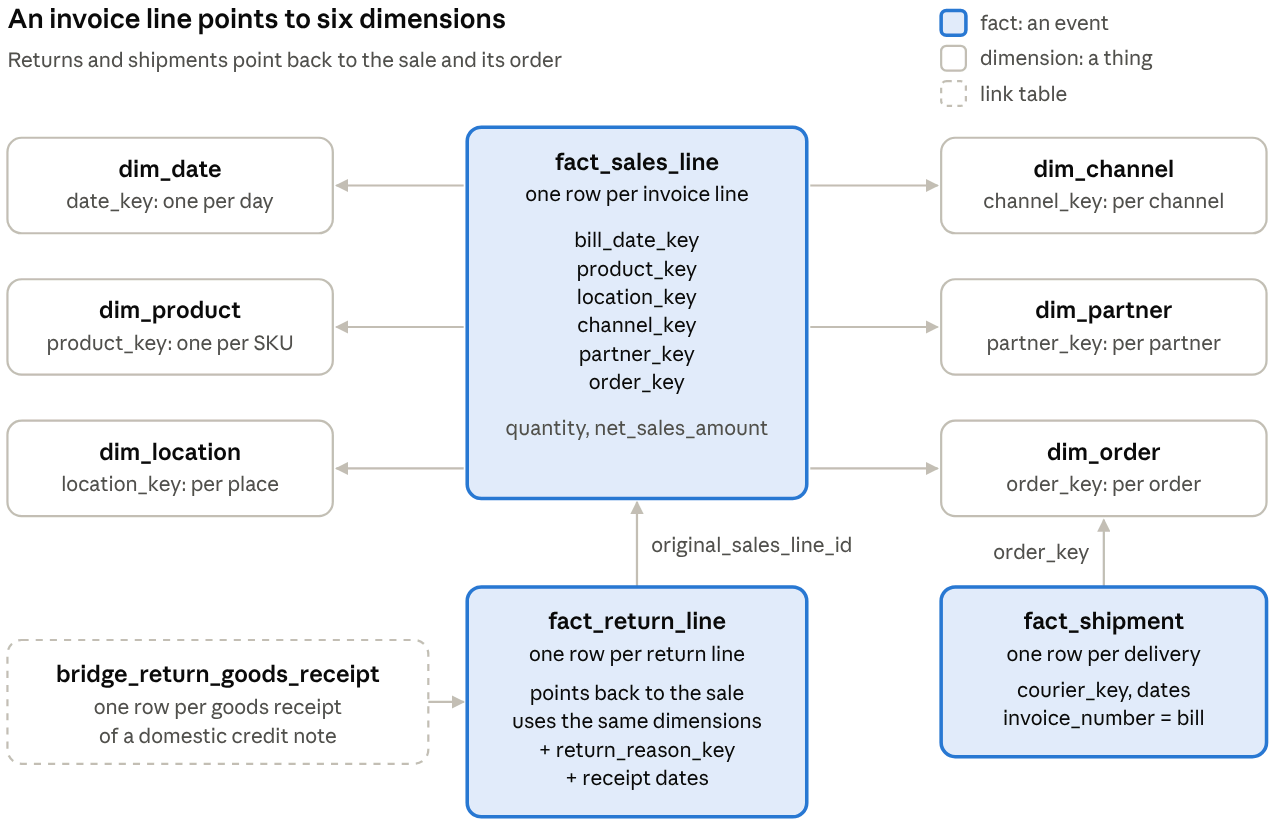
Table	One row per	Key	Example	Comes from	Rows (sample)
dim_date	Calendar day	date_key (YYYYMMDD)	20260915 = Tue 15 Sep 2026, FY 2026-27, Q2	Generated	2,556
dim_product	Item code (SKU, packaging, charge line)	product_key	GSTEA2724, Glass Bottle, Trading, Continued, MRP 850	Final Master, stock report, invoices, POs	4,465
dim_product_history	Version of a product's status, MRP, category	product_key + valid_from	MRP 800 until 1 Sep, 850 from 1 Sep	Each new Final Master	one per change
bridge_product_component	Component of a set	set + component product keys	"Set of 2" = 2 x the single item	Product team (gap G-010)	0 so far
dim_location	Place stock can sit	location_key	BAGRU_WH, an Amazon FBA, a store, Blinkit consignment, "With employees"	Reviewed location mapping	77
dim_channel	Sales channel and sub-channel	channel_key	Amazon / FBA, Own Website / Shopify, Franchise	Reviewed channel mapping	23
dim_partner	Counterparty name in a role	partner_key	Shoppers Stop Limited (customer); a factory (supplier)	Every name in the files, classified by the partner rules	253
dim_order	Customer order	order_key	Shopify order 124136093; Amazon 405-6175875-6064319	Order numbers on invoices and returns	25,856
dim_courier	Courier	courier_key	Delhivery	Invoices, later the warehouse export	6
dim_return_reason	Reason for a return	return_reason_key	Only "Not recorded" today	Gap G-009	1
dim_movement_type	Reason stock moved	movement_type_key	SALE, TRANSFER_OUT, CONSIGNMENT_IN…	Defined in the design	18

Two dimensions need a word more:

- **dim_partner** carries the classification that decides how a sale is treated: B2B (final), franchise, SIS or quick commerce (sale-or-return), group export (at cost), employee (not a sale), or UNCLASSIFIED. It also records how the classification was made (listed by the business, or by a default rule) and which rule it follows.

- **dim_location** includes two virtual locations that exist only in the design: "Blinkit (Kiranakart) consignment stock" and "With employees (exhibitions)". They let the ledger track stock that has left our warehouses but is still ours.

4. Sales and returns



sales and returns · one invoice line, six dimensions, returns and shipments

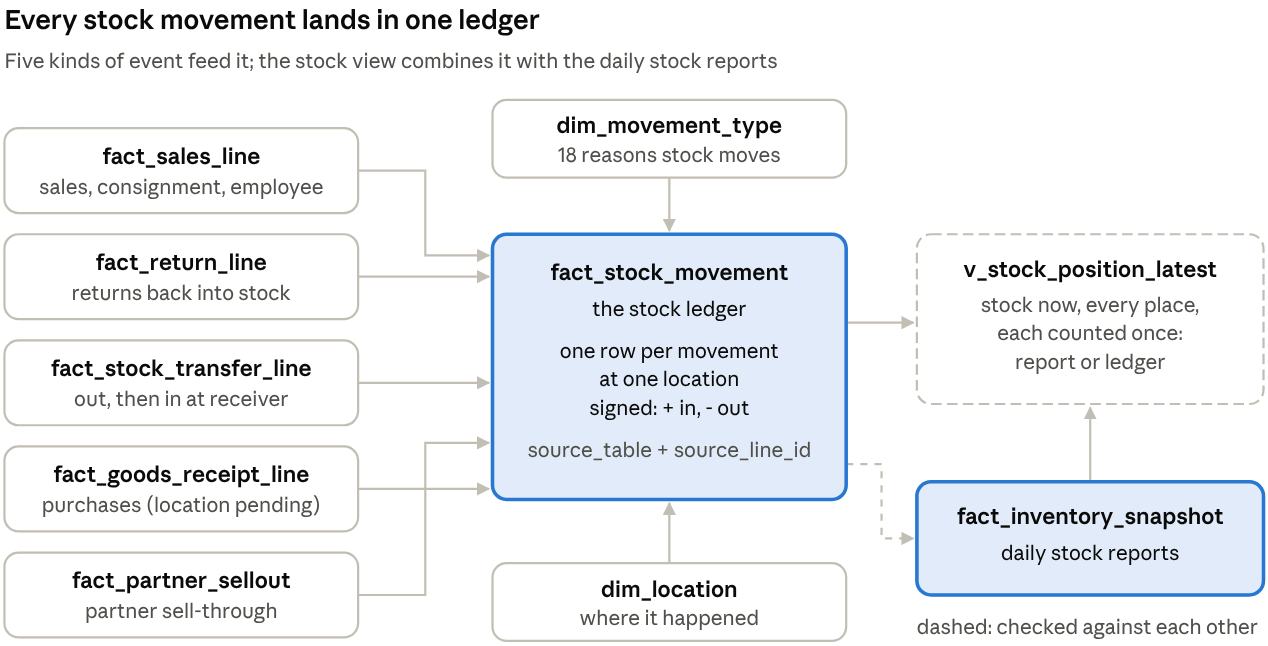
Table	One row per	Connects to	Rows (sample)
fact_sales_line	Invoice line (Retail and Domestic Sales)	dim_date, dim_product, dim_location, dim_channel, dim_partner, dim_order	31,150

Table	One row per	Connects to	Rows (sample)
fact_return_line	Return or credit-note line	The same dimensions, plus dim_return_reason; `original_sales_line_id` points at the invoice line it reverses	18,788
bridge_return_goods_receipt	Goods receipt behind a domestic credit note	fact_return_line through `return_line_id`	8,705
fact_shipment	Delivery from the warehouse	dim_order, dim_channel, dim_location, dim_courier; `invoice_number` matches `fact_sales_line.bill_number`	0 (export not received yet)

How to read it:

- **Net sales = invoices minus returns**, both excluding GST. The view `mart.v_net_sales_daily` does this subtraction for you.
- **A return knows which sale it reverses** when that sale is in the loaded data. Returns of sales made before the data starts have no link, and that's expected.
- **An order ties everything together.** One website or marketplace order can have several invoice lines, a shipment and a return; they all share the same `order_key`. B2B invoices have no channel order, so they use key -1.
- **Domestic credit notes can cover several goods receipts on different dates**, so a return line keeps two kinds of date: the credit-note date (for net sales) and the first and last goods-receipt dates (for when the stock physically came back). The bridge lists each receipt.

5. Stock: reports, transfers and the ledger



stock · five event tables feed the ledger; the stock view uses reports and the ledger

There are two ways to know how much stock there is, and the warehouse keeps both.

- **The stock reports** (`fact_inventory_snapshot`) say what each system counted on a given day: one row per day, location and SKU, non-zero rows only. They come from the ERP stock report, Blinkit and Instamart. Past days can't be re-exported, so history builds up from the day reports are saved.
- **The stock ledger** (`fact_stock_movement`) records every movement as it happened: one row per movement at one location, with a plus or minus sign. A transfer from Bagru to an FBA produces two rows: minus at Bagru when it leaves, plus at the FBA when it arrives. Between those dates the stock is in transit.

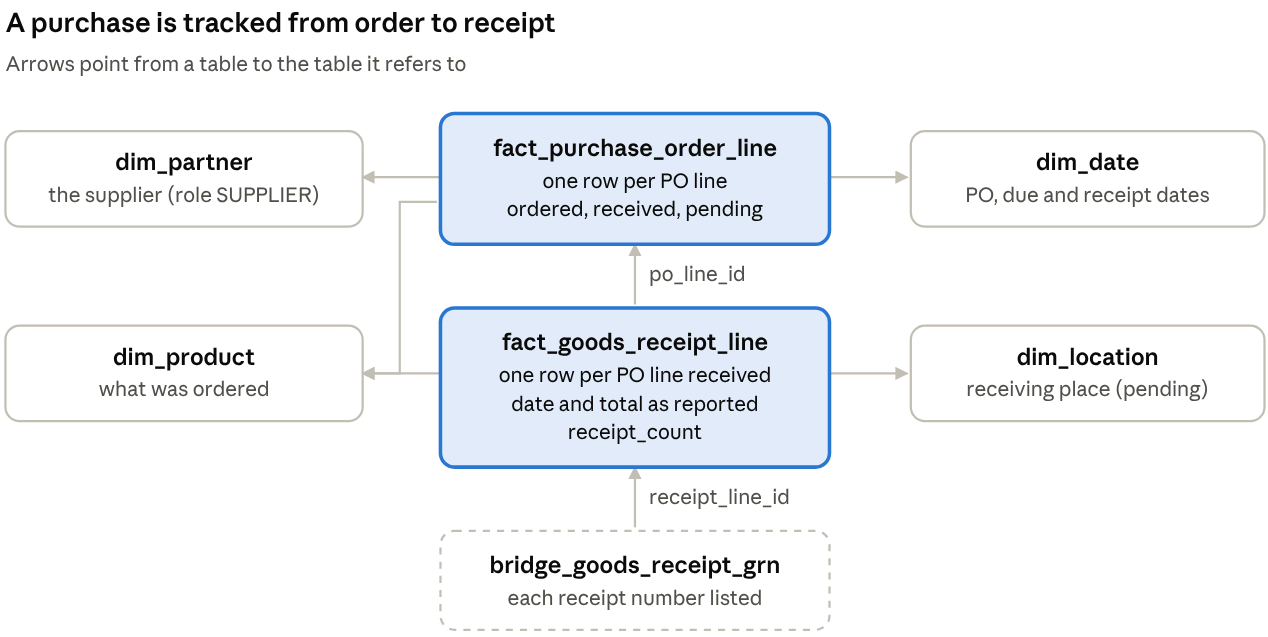
Each ledger row says why the stock moved (`movement_type_key`) and which event created it (`source_table` + `source_line_id`), so every movement can be traced to its invoice, return, transfer or receipt line.

Event	Ledger rows it creates
Sale to a customer or an outright B2B buyer	SALE: minus at the selling location

Event	Ledger rows it creates
Invoice to a franchise, SIS or quick-commerce partner	CONSIGNMENT_OUT minus at the warehouse, CONSIGNMENT_IN plus at the partner's location: still our stock (rule BR-022)
Stock taken by an employee for an exhibition	TRANSFER_OUT minus at the warehouse, TRANSFER_IN plus at "With employees" (rule BR-029)
Invoice to a not-yet-classified partner	ISSUE_UNCLASSIFIED: minus at the warehouse, destination unknown, never guessed
Customer or partner return	SALE_RETURN, or the consignment / employee return pair, dated when the goods came back
Stock transfer	TRANSFER_OUT on the sending date, TRANSFER_IN on the receiving date
Goods receipt from a factory	GOODS_RECEIPT plus, once the receiving location is in the export (gap G-025)

The view `mart.v_stock_position_latest` counts each location once: from its own stock report where one exists (warehouses, stores, FBA, Blinkit, Instamart), and from the ledger where none does (partner consignment, employees). The two are checked against each other: today's report minus all recorded movements should never imply negative opening stock; where it does, an inflow is missing from the ledger, and a quality check reports it.

6. Purchasing



purchasing · PO line, its receipt row, and the receipt numbers

Table	One row per	Connects to	Rows (sample)
fact_purchase_order_line	PO line, in its latest state	dim_partner (supplier), dim_product, dim_date (PO and due dates)	1,273
fact_goods_receipt_line	PO line that has received goods	The PO line (`po_line_id`), supplier, product, receipt date, receiving location	753
bridge_goods_receipt_grn	Receipt number on a PO line	The receipt row (`receipt_line_id`)	819

How to read it:

- **The PO table is a state, not a history.** Each new PO export replaces it with the latest status of every line (open, in progress, overdue, received, closed, cancelled). Older exports stay in raw.

- **Receipts are stored exactly as the PO report gives them.** A line delivered in several parts lists all its receipt numbers in one cell, with one date and one total quantity. So the receipt row holds the total and that one date, `receipt_count` says how many receipts there were, and the bridge lists each number. Lead times for those lines are marked approximate until a per-receipt register is exported (gap G-027).
- **Not every PO counterparty is a supplier.** Customers and own stores sending stock back also appear on this report. Until the factory list arrives, these counterparties stay unclassified, and their receipts are not yet placed in the stock ledger.
- **The receiving location is not in the export yet** (gap G-025), so receipts carry location -1 for now.

7. What the agents write

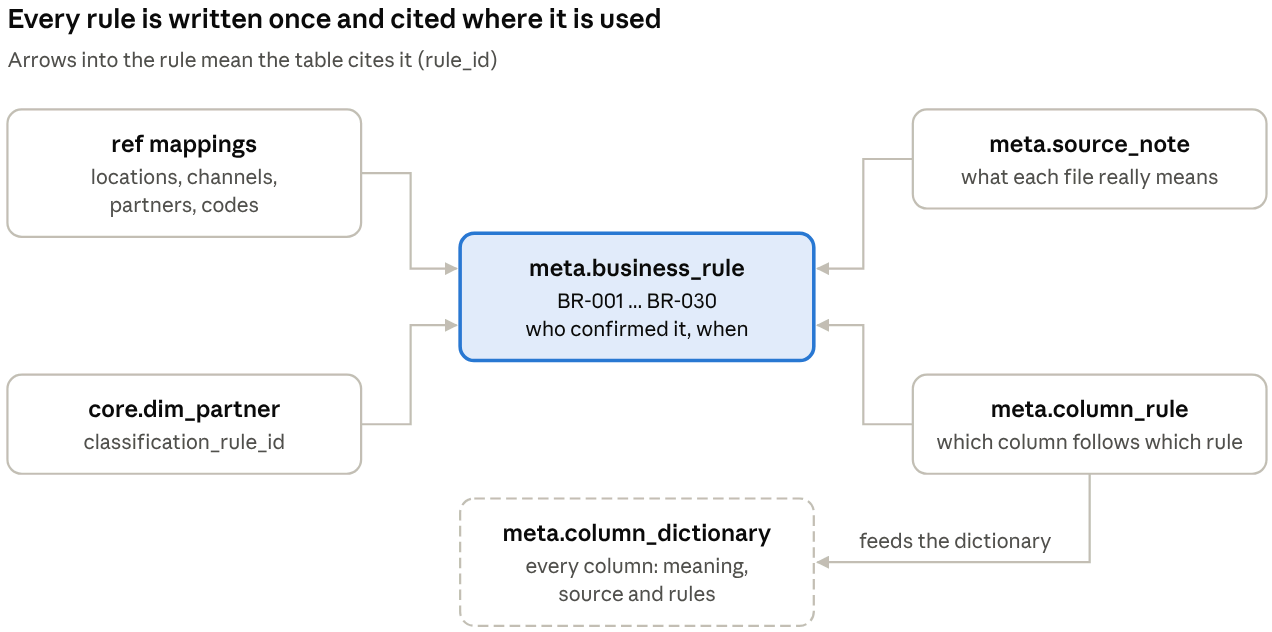
The agents read from **mart** and **core**, and write what they produce to the **insight** schema. Every piece of advice is kept, together with what a person decided about it, so the agents' accuracy can be measured over time (rule BR-028). These tables are empty until the agents are built.

Table	One row per	Connects to	Example
insight.model_run	Run of an agent	Everything below points to it (`run_id`)	Stock AI demand forecast, model version 0.1, data as of 15 Sep 2026
insight.demand_forecast	Product x location x channel x period	dim_product, dim_location, dim_channel	12 units (range 8-16) of GSTEA2724 at an FBA next week
insight.recommendation	Suggested action	dim_product, from and to dim_location, supplier in dim_partner	Move 12 units Bagru to FBA because cover is under 7 days; accepted by the ops lead on 16 Sep
insight.alert_rule	Kind of alert	Referenced by alert_event	Low stock: cover under 7 days, warning, to the ops team

Table	One row per	Connects to	Example
insight.alert_event	Alert raised	alert_rule, and the product, location or channel it concerns	Cover at FBA DED4 fell to 3 days; acknowledged

A recommendation can't be marked accepted or rejected without saying who decided and when; the database refuses it otherwise.

8. The rulebook and the audit trail



rulebook · business rules cited by mappings, partners, notes and the dictionary

Everything we know about the data that isn't data itself is stored in **ref** and **meta**, so it never lives only in someone's head.

The rulebook answers "why is this number what it is?"

Table	Holds
meta.business_rule	30 numbered rules, e.g. BR-001 "headline sales exclude GST", BR-007 "exports go through Dileep at cost". Each says who confirmed it and when. Rules are never deleted, only retired or replaced.
ref.location_master, ref.location_alias	Every location, and every spelling of it in every file
ref.channel_master, ref.channel_alias	Every channel, and the values in each file that map to it
ref.partner_type, ref.partner_list, ref.partner_default_rule	How counterparties are classified, and the partners listed by name
ref.code_mapping	Fixed translations: PO statuses, product-type words, which codes are charges
meta.source_note	What only the person handling a file knows: column meanings, traps, export quirks
meta.column_dictionary	One row per warehouse column: its meaning, the file and column it comes from, and its rules

The audit trail answers "where did this row come from, and can I trust it?"

Table	Holds
meta.source_registry	The 14 kinds of file the warehouse accepts
meta.load_batch	One row per file loaded: name, fingerprint, period, row count, status. Every row in raw and core carries its `batch_id`.
meta.transform_run	One row per build, with its period and outcome
meta.dq_check_result	Every quality check of every load and build
meta.business_process, meta.process_coverage	The 24 business processes, and which months each has data for

Table	Holds
meta.data_gap	28 known gaps between the design and today's data, with owners

9. How to join the tables

A join always matches a key in one table to the same key in another. Every connection in the warehouse is listed here.

From	Column	To	Column
Any fact	*_date_key (bill, return, document, po, grn, snapshot, movement)	dim_date	date_key
Any fact	product_key	dim_product	product_key
fact_sales_line, fact_inventory_snapshot, fact_stock_movement	location_key	dim_location	location_key
fact_return_line	receiving_location_key	dim_location	location_key
fact_stock_transfer_line	from_location_key, to_location_key	dim_location	location_key
fact_sales_line, fact_return_line, fact_partner_sellout, fact_shipment	channel_key	dim_channel	channel_key
Any fact with a counterparty	partner_key	dim_partner	partner_key
fact_sales_line, fact_return_line, fact_shipment	order_key	dim_order	order_key
fact_return_line	original_sales_line_id	fact_sales_line	sales_line_id
fact_shipment	invoice_number	fact_sales_line	bill_number
fact_stock_transfer_line (IN)	linked_outward_document	fact_stock_transfer_line (OUT)	document_number
fact_goods_receipt_line	po_line_id	fact_purchase_order_line	po_line_id
bridge_goods_receipt_grn	receipt_line_id	fact_goods_receipt_line	receipt_line_id
bridge_return_goods_receipt	return_line_id	fact_return_line	return_line_id
fact_stock_movement	movement_type_key	dim_movement_type	movement_type_key
fact_stock_movement	source_table + source_line_id	The fact named in source_table	Its line id
Any raw or core fact row	batch_id	meta.load_batch	batch_id

Example 1: net sales by channel in August 2026

The easy way is the ready-made view, which already subtracts returns and leaves out employee stock:

```
SELECT channel_name, sum(net_sales) AS net_sales
FROM mart.v_net_sales_daily
WHERE month_start_date = DATE '2026-08-01'
GROUP BY channel_name
ORDER BY net_sales DESC;
```

The same from core, invoices only (before returns), shows how the joins work:

```
SELECT c.channel_name, sum(s.net_sales_amount) AS invoiced
FROM core.fact_sales_line s
JOIN core.dim_channel c ON c.channel_key = s.channel_key
JOIN core.dim_date d    ON d.date_key    = s.bill_date_key
WHERE d.month_start_date = DATE '2026-08-01'
GROUP BY c.channel_name
ORDER BY invoiced DESC;
```

In the sample, the website shows Rs 42.6 lakh invoiced but Rs 39.0 lakh net. The difference is the August returns.

Example 2: which sale does each return reverse?

```
SELECT r.return_number, s.bill_number, p.sku_code, r.quantity
FROM core.fact_return_line r
JOIN core.fact_sales_line s ON s.sales_line_id = r.original_sales_line_id
JOIN core.dim_product p     ON p.product_key   = r.product_key;
```

Example 3: the stock story of one SKU at Bagru

```
SELECT d.calendar_date, t.movement_name, m.quantity
FROM core.fact_stock_movement m
JOIN core.dim_movement_type t ON t.movement_type_key = m.movement_type_key
JOIN core.dim_location l      ON l.location_key      = m.location_key
JOIN core.dim_product p       ON p.product_key       = m.product_key
JOIN core.dim_date d          ON d.date_key          = m.movement_date_key
WHERE l.location_code = 'BAGRU_WH' AND p.sku_code = 'GSTEA2724'
ORDER BY d.calendar_date;
```

Each row is one movement: sales out, returns back, transfers to FBAs. Adding up `quantity` gives the net change over the period.

10. Glossary

Term	Meaning
Fact	A table of events (sales, returns, movements). Grows every day; holds quantities and amounts.
Dimension	A table that describes things (product, location, channel, partner, date). Changes slowly.
Key	A number that identifies one row of a dimension, e.g. product_key. Facts store keys instead of names.
Join	Matching a key in one table to the same key in another, to look up names and attributes.
Grain	What one row of a table represents, e.g. "one invoice line". Mixing grains double-counts.
-1 (Unknown)	The key used when a value in a file matches nothing. The row is kept and counted, never dropped.
source_sku_code	The SKU exactly as written in the file, kept next to product_key for tracing.
line_occurrence	1 for the first line with a given document and SKU, 2 for a repeat line on the same document.
batch_id	The file load a row came from (meta.load_batch).
Bridge (link table)	A table that lists several items belonging to one row, e.g. the receipt numbers of one PO line.
Ledger	fact_stock_movement: every stock movement as a signed row, plus in and minus out.
Snapshot	A stock report for one day. Can't be re-exported later, so saved daily from now on.

Term	Meaning
System of record	The report treated as the truth for a location's stock, so each location is counted once.
Sale-or-return / consignment	Stock sent to franchise, SIS and quick-commerce partners stays ours until they sell it or send it back.
UNCLASSIFIED	A counterparty not yet on the reviewed partner list. Kept visible, never guessed (safety switch).
Net sales	Invoices minus returns, excluding GST (rules BR-001 to BR-004).
Business rule (BR-nnn)	A numbered, confirmed statement about how the business or the data works.
Gap (G-nnn)	Something the design needs that today's data doesn't provide, with an owner and how to fill it.
Cut-off	The last business date included (15 Sep 2026 for the prototype). A setting, not code.
Quality gate	Checks that must pass for a build to be published; otherwise the whole build is rolled back.